Species distribution modelling with Poisson point processes

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Species distribution modelling

Envrionmental data

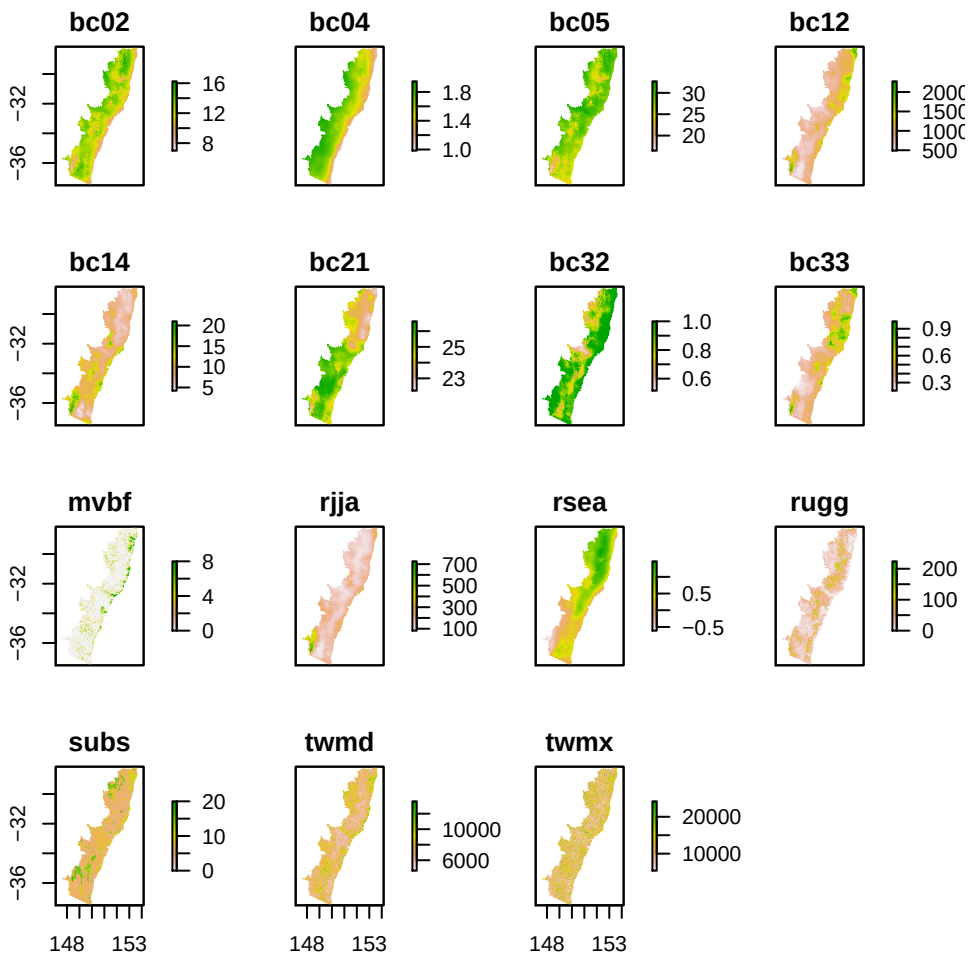
```
# library(maptools)
library(raster)
library(sf)

grid_dir <- "data/grids"
vars <- list.files(grid_dir, pattern = ".tif$", full.names = TRUE)

# the border shapefile
ibra <- st_read("data/ibraone.shp", crs = 4283, quiet = TRUE)

# read the raster layers as a raster stack
nsw_stack <- stack(vars)
# set the coordinate system
crs(nsw_stack) <- CRS("+init=epsg:4283")

plot(nsw_stack)
```



Species data

The species data comes form Fithian and others (2015).

All the species starts with `euca` . . .

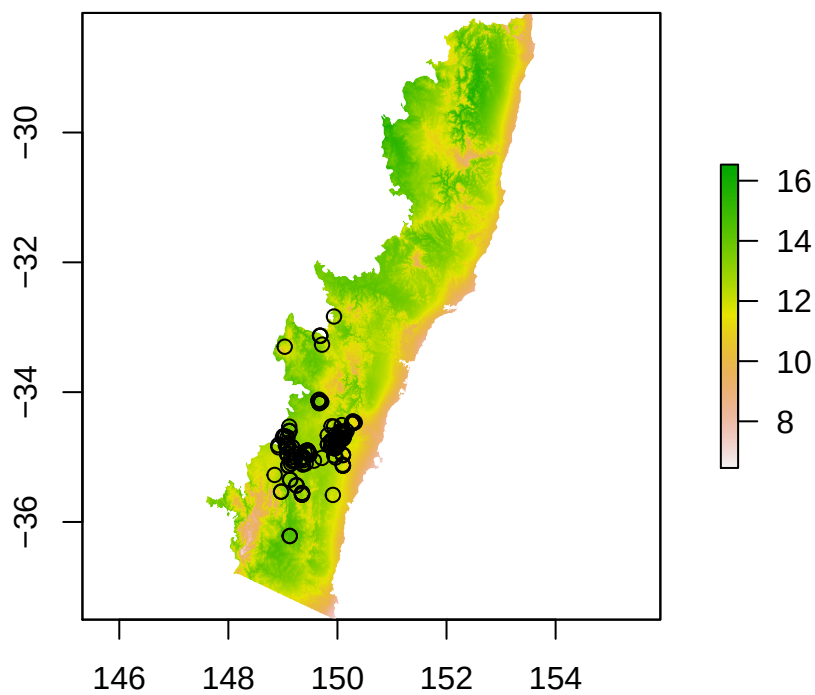
```
# load Fithian et al (2015) species data
load("data/moddat.RData")

head(eucacine_moddat)
```

```
##   id     long       lat year accuracy  longlat   cells     bc01     bc02
## 1  1 150.0862 -34.60553 2000      100 15008622 6279811 13.63983 12.23118
## 2  2 150.2854 -34.44967 2000      100 15028535 6126927 13.24499 11.36682
## 3  3 150.0125 -34.80654 2000      100 15001250 6474021 13.71813 12.41187
## 4  4 150.0997 -34.96775 2000      100 15009972 6631876 13.09869 11.85747
## 5  5 149.8980 -34.85114 2007      100 14989800 6517680 13.29461 12.52573
## 6  6 149.7147 -35.01311 1999      100 14971468 6675426 12.81009 12.73738
##       bc04     bc05        bc06     bc12      bc14     bc21      bc28      bc32
## 1 1.645262 27.27752  1.09317458 693.4518  9.081451 26.44496 0.5211285 0.7648971
## 2 1.596420 26.22510  1.17330492 780.2014 10.506531 25.97984 0.6106081 0.8590701
## 3 1.652387 27.46763  1.06588304 681.5583  9.605835 26.38567 0.5021659 0.7421684
## 4 1.613784 26.41990  0.87655205 788.7050 10.496765 25.96847 0.5949574 0.8564287
```

```
## 5 1.699777 27.45464  0.58909386 653.2943  9.165375 26.40723 0.4897964 0.7353666
## 6 1.749456 27.40740 -0.01114328 677.7914 10.394196 26.34236 0.5267761 0.8032433
##        bc33      d2r      d2t     elev ibgd ibsb mvbf     rdjf     rjja
## 1 0.3000291   0.0000 76729.03 590.4653   98  394    3 188.9394 144.8310
## 2 0.3601428   0.0000 55561.12 692.9059   93  384    0 214.0642 160.9149
## 3 0.2900780   0.0000 91516.44 550.2057   98  394    0 183.8921 140.0840
## 4 0.3360529   0.0000 91472.76 639.3138   98  394    0 206.9004 166.6616
## 5 0.2770475 742.9186 84018.19 623.7888   98  394    0 175.7002 134.8407
## 6 0.2836685   0.0000 59406.70 696.1747   98  392    3 177.5476 147.8208
##        rsea      rugg sandsubs     slpe stre subs twmd  twmx      wpot xveg
## 1 0.2742345 10.927129        0 1.216894 0.76    5 6260  7542 -1.288309    0
## 2 0.2579768 18.860123        0 3.167662 0.00    7 8023 11071 -1.171590    0
## 3 0.2861401 19.935825        0 1.852355 0.00    5 7474  9066 -1.308515    1
## 4 0.2572912 15.468473        0 2.161766 0.00    5 7278  9110 -1.118788    1
## 5 0.2735702  8.366111        0 1.390608 1.00   15 8569 13911 -1.360653    0
## 6 0.2330509  5.964362        0 1.133822 0.00    6 8172 10061 -1.218756    1
```

```r
plot(nsw_stack[[1]])
points(eucacine_moddat$long, eucacine_moddat$lat)
```



Modelling with Maxent

Modelling with Poisson point processes

Comparing the results